

Ethan Gray Mechatronics Engineering Student

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TECHNICAL SKILLS

Software

Python, OpenGL, Pandas, NumPy, C, C++, CMSIS-DSP, MATLAB, Java

Electrical

Altium, NI Multisim, Embedded Programming, Reading Datasheets

Mechanical

SolidWorks, Ansys, 3D Printing, Laser/Waterjet Cutting, Machining

TECHNICAL EXPERIENCE

Product Design Mechanical Engineering Intern, Verkada

01/2025 – 08/2025

- Led mechanical design of two motion sensors, a zone expander, and an access control relay, bringing them from initial product requirements and competitor benchmarking to tooling kick-off
- Implemented high volume DFMA for plastic injection molding, compression molding, die cutting, and PCB design
- Optimized optical and microwave design for motion sensors, extending detection range while preserving slim profile
- Collaborated with our vendor teams in Taiwan concerning mechanical design, PCB layout compatibility, and validation

Avionics, UBC AeroDesign

05/2023 – 05/2025

- Developed a compartmentalized toolbox for GNSS-aided inertial navigation in Python to provide accurate position, velocity, and attitude estimates to other software modules, and later ported it to C with CMSIS-DSP to run on our MCU
- Utilized OpenGL, Pygame, Folium, NumPy, and Matplotlib to support validation efforts by visualizing flight data
- Wrote sensor drivers and firmware to run on an STM32 for collecting and packaging sensor data for navigation

Advanced Fuselage, UBC AeroDesign

09/2021 – 05/2023

- Engineered and validated lightweight tail boom, empennage, and sensor mount attachments in SolidWorks
- Manufactured the fuselage using laser cutting, waterjet cutting, 3D printing, carbon fiber layups, and machining

Mechanical Engineering Intern, Seaspan ULC

09/2023 – 12/2023

- Modified Joint Support Ship piping design across 6 zones, compiling detailed packages for implementation in the build
- Led proof-of-concept and financial analysis that supported purchase of machinery that saves \$200K per vessel

Field Engineering Intern, EBC

05/2023 – 08/2023

- Oversaw concrete operations in the powerhouse for the Site C Dam by managing pours and the subsequent cooling
- Managed inspections with BC Hydro engineers to verify formwork, rebar, grounding cables, and second stage inserts

TECHNICAL PROJECTS

AllSpark, C++, Python, asyncio, WebSockets, SolidWorks, 3D Printing, Altium, Circuit Design

- Designed a piezoresistive force measurement unit that uses WebSockets for wireless data transmission to a host
- Designed a 4-layer PCB in Altium around an ESP32-C3-MINI-1-N4, featuring optimized AFE and power multiplexing
- Developed ESP32 firmware for sensor data acquisition and transmission, and host-side software for calibration, processing, visualization, and multi-client support
- Engineered an assembly in SolidWorks rated for 100 pounds per unit, optimized for manufacturability and assembly

Night Vision Monocular, C, MMAL, Bash, SolidWorks, 3D Printing, Altium, Circuit Design

- Architected and manufactured a digital night vision monocular that is compatible with helmet mounting technology and is competitive in cost and performance with available alternatives
- Developed software for camera interfacing using MMAL API and displayed the picture on an LCD module at 60fps
- Built a circuit to power the Raspberry Pi with a 3.7V lithium battery and power budgeted for 5 hours of battery life
- Modeled and iterated SolidWorks housing across three revisions, improving electronic storage, assembly, and mounting

Discord Bot, Python, BS4, discord.py, NumPy, Matplotlib

- Programmed a Discord bot that displays current prices of in-game items and graphs price history in Discord embeds
- Implemented web scraping with BS4 to populate a custom data repository that updates automatically on an hourly basis

EDUCATION

University of British Columbia, BAsC – Mechatronics Engineering, CGPA: 91.3%

2021 – 2026